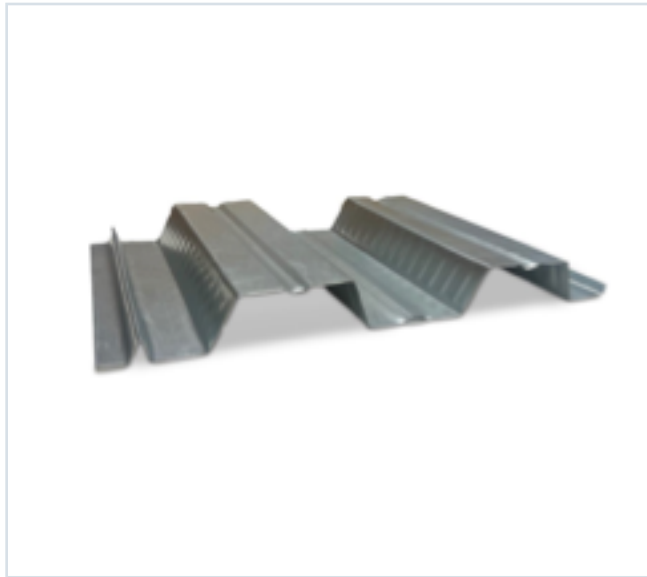


DECKING PRODUCT SUBMITTAL

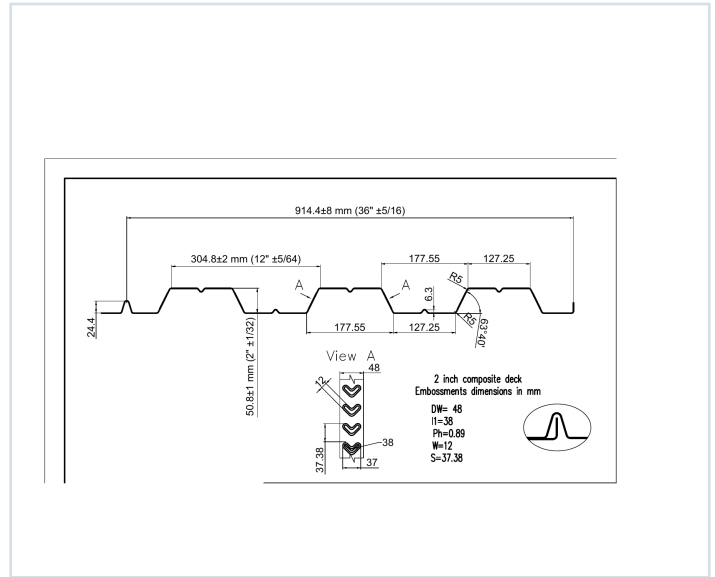
2CD-FY50-18-G60

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	18 ga	G60

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.5	50	0.493	0.500	14760	14960	0.548	0.530

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

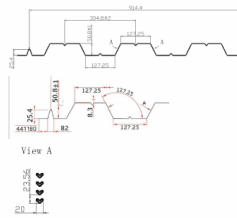
2CD-FY50-18-G60 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E, ksi	S _x (in ³) per foot	S _y (in ³) per foot	ASD (I _x + I _y)		I _x + I _y (in ⁴) per ft.	I _x + I _y (in ⁴) per ft.
						M _x / 12 in ³ -ft. per ft.	M _y / 12 in ³ -ft. per ft.		
22	0.0295	16	50	0.247	0.254	7407	7595	0.381	0.290
20	0.0358	19	50	0.329	0.336	9860	10060	0.393	0.373
18	0.0474	25	50	0.483	0.500	14760	14960	0.548	0.530
16	0.0598	32	50	0.646	0.644	19321	19271	0.703	0.683

Note: All section properties and ASD based strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10, and AISC 360-10.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _w) (kips)		Web Crippling (R _w) (kips)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	579
20	278	295	327	364	398
18	346	366	403	446	490
16	440	467	517	571	625

Note: All section properties and ASD based strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10, and AISC 360-10.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	178	171	164	157	150	143	136	129	122	115
	18	273	267	258	249	240	231	222	213	204	195	186
	16	358	351	340	329	318	307	296	285	274	263	252
Double	22	131	125	119	113	107	101	95	89	83	77	71
	20	186	179	172	165	158	151	144	137	130	123	116
	18	277	269	260	251	242	233	224	215	206	197	188
	16	357	349	338	327	316	305	294	283	272	261	250
Triple	22	176	169	162	155	148	141	134	127	120	113	106
	20	243	235	227	219	211	203	195	187	179	171	163
	18	346	337	328	319	310	301	292	283	274	265	256
	16	446	436	426	416	406	396	386	376	366	356	346

Note: 1. All section properties and ASD (I_x + I_y) uniform loads are calculated in accordance with ANSI/AISC 360-10, AISC 360-10, and AISC 360-10.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for fixed-end moments are: Single and Two-Span: $M = \frac{wL^2}{8}$; Three-Span: $M = \frac{wL^2}{16}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	13	13	12	11	10	9	8	7	6	5	4
	18	13	13	12	11	10	9	8	7	6	5	4
	16	13	13	12	11	10	9	8	7	6	5	4
Double	22	20	19	18	17	16	15	14	13	12	11	10
	20	20	19	18	17	16	15	14	13	12	11	10
	18	20	19	18	17	16	15	14	13	12	11	10
	16	20	19	18	17	16	15	14	13	12	11	10
Triple	22	24	23	22	21	20	19	18	17	16	15	14
	20	24	23	22	21	20	19	18	17	16	15	14
	18	24	23	22	21	20	19	18	17	16	15	14
	16	24	23	22	21	20	19	18	17	16	15	14

Note: For loads that cause L/10 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-18-G60 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-8"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-8"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-3"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-8"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-8"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-8"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-8"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	298	290	280	276	260	247	236
5	304	295	290	277	268	255	240	230
5.5	304	300	291	285	272	255	240	230
6	307	307	304	304	298	285	270	260
6.5	307	307	304	304	304	298	285	270

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	256	250	236	220	204	190	170
4.5	267	267	261	246	230	214	200	180
5	277	277	271	254	238	222	208	188
5.5	284	284	278	260	244	228	214	194
6	290	290	284	266	250	234	220	200
6.5	290	290	284	266	250	234	220	200